

Intelligent Media Processing Pipeline

Introduction

The Intelligent Media Processing Pipeline is a fully containerized microservices-based application designed to automate the analysis of vehicle images through advanced AI techniques. This system provides a complete workflow from image upload through to intelligent analysis, including blur detection, brightness assessment, duplicate identification, and Optical Character Recognition (OCR) for vehicle number plate validation.

As demonstrated across the system's components, this pipeline integrates a React-based frontend, Node.js backend, MongoDB for metadata storage, Redis for caching and queue management, and a dedicated worker service for asynchronous processing. The architecture ensures scalable, efficient, and real-time image analysis with a user-friendly interface for uploading, monitoring, and reviewing results.

Technology Stack Used

Layer	Technology	Version	Purpose
Runtime	Node.js	v18+	JavaScript runtime
Backend Framework	Express.js	v4.18+	REST API
Database	MongoDB	v6.x	Document storage
ODM	Mongoose	v7.x	Schema validation
Cache/Queue	Redis	v7.x	In-memory cache
Queue Manager	BullMQ	v4.16+	Job processing
Image Processing	Sharp	v0.32+	Image manipulation
OCR	Tesseract.js	v4.x	Text extraction
File Upload	Multer	v1.4+	Multipart handling
Frontend Framework	React	v18.x	UI components
Build Tool	Vite	v4.x	Development/build
HTTP Client	Axios	v1.x	API requests
Testing	Jest	v29.x	Unit/integration tests
Containerization	Docker	v24.x	Container management
Orchestration	Docker Compose	v2.x	Multi-container setup

Screenshot

Frontend:

1)

☐		Name	Container ID	Image	Port(s)	CPU (%)	Memory usag...	Memory (%)	Disk	Actions
☐	▼	intelligent-media-pipe	-	-	-	1.67%	387.42MB / 37.1	5.09%	28	
☐		media-redis	d281eaa9b0fc	redis:7-alpi	6379:6379	0.43%	6.05MB / 7.43GB	0.08%	0B	
☐		media-mongodb	ddb2d6c856e	mongo:6	27017:27017	0.98%	169.2MB / 7.43G	2.22%	28	
☐		media-worker	2445e6ce756d	intelligent-i	-	0%	58.66MB / 7.43G	0.77%	0B	
☐		media-backend	0db0b2fa5ab0	intelligent-i	5000:5000	0.11%	70.5MB / 7.43GB	0.93%	0B	
☐		media-frontend	e5f43ed4f7be	intelligent-i	5173:5173	0.15%	83.01MB / 7.43G	1.09%	0B	

2)

Intelligent Media Processing Pipeline
Upload vehicle images for AI-powered analysis

Upload Image

Select an image to begin processing

Drop image here or click to select

JPEG, PNG, WEBP (max 10MB)

Upload Image

Recent Uploads

No images uploaded yet

Processing History

No history yet

Quick Actions

New Upload

Refresh

Update Stats

Help & Tips

Supported formats: JPEG, PNG, WEBP

Max file size: 10MB

Processing typically takes 5-10 seconds

Results are heuristic and probabilistic

OCR accuracy depends on image quality

3)



Intelligent Media Processing Pipeline

Upload vehicle images for AI-powered analysis

Processing Status

[New Upload](#)

Processing Status

ID: `img_1786277811294_chs9ot`

 Initializing...

 Polling... Attempt 0

Recent Uploads

No images uploaded yet

Processing History

 `img_1786277...` 8/9/2026, 5:46:51 PM

Quick Actions

[New Upload](#)

[Refresh](#)

[Update Stats](#)

Help & Tips

-  Supported formats: JPEG, PNG, WEBP
-  Max file size: 10MB
-  Processing typically takes 5-10 seconds
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-  OCR accuracy depends on image quality

4)



Intelligent Media Processing Pipeline

Upload vehicle images for AI-powered analysis

Analysis Results

[New Upload](#)

Analysis Results

Image Metadata

Dimensions: 960 x 1280
Format: jpeg
Size: 400.1 KB

Blur Detection

Status:  Image is clear
Score: 0 (Threshold: 100)
Note: This is a heuristic measurement

Brightness Analysis

Status:  Image is clear
Average Brightness: 0
Note: This is a heuristic measurement

Duplicate Detection

Status:  No duplicates found
Similarity Score: 0
Note: Perceptual hashing may have false positives

OCR Results

Extracted Text: None detected
Confidence: 0%

Vehicle Number Validation

Normalized: None
Valid Format:  No
Note: Format validation only; not government verification

 All analysis results are heuristic and probabilistic. They should be used as indicators, not definitive proof.

[Upload Another Image](#)

Recent Uploads

No images uploaded yet

Processing History

 `img_1786277...` 8/9/2026, 5:46:51 PM

Quick Actions

[New Upload](#)

[Refresh](#)

[Update Stats](#)

Help & Tips

-  Supported formats: JPEG, PNG, WEBP
-  Max file size: 10MB
-  Processing typically takes 5-10 seconds
-  Results are heuristic and probabilistic
-  OCR accuracy depends on image quality

Backend:

5)

```
{
  "name": "Intelligent Media Processing Pipeline API",
  "version": "1.0.0",
  "status": "running",
  "endpoints": {
    "upload": "POST /api/images/upload",
    "status": "GET /api/images/:id/status",
    "results": "GET /api/images/:id/results",
    "retry": "POST /api/images/:id/retry",
    "list": "GET /api/images",
    "stats": "GET /api/images/stats",
    "health": "GET /health"
  },
  "documentation": "See README.md for more information"
}
```

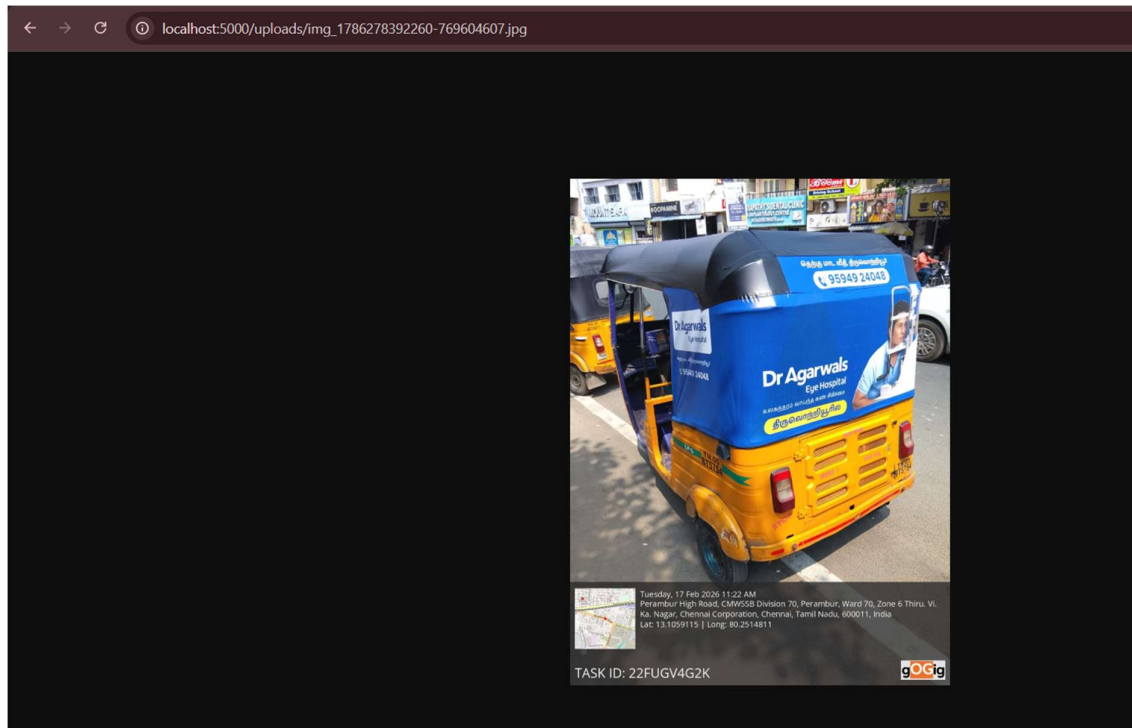
6)

```
C:\Users\gokul\Downloads\intelligent-media-pipeline\backend>curl -X POST http://localhost:5000/api/images/upload -F "image=@53916.jpg"
{"success":true,"processingId":"img_1786278392306_jl21tb","status":"pending","message":"Image uploaded successfully"}
```

7)

```
{
  "success": true,
  "count": 22,
  "path": "/app/uploads",
  "files": [
    {
      "name": "img_1786271701273-89099447.jpg",
      "url": "/uploads/img_1786271701273-89099447.jpg",
      "size": 61664,
      "modified": "2026-08-09T10:35:01.276Z"
    },
    {
      "name": "img_1786272724673-287498971.jpg",
      "url": "/uploads/img_1786272724673-287498971.jpg",
      "size": 26474,
      "modified": "2026-08-09T10:52:04.686Z"
    },
    {
      "name": "img_1786274518473-60644524.jpeg",
      "url": "/uploads/img_1786274518473-60644524.jpeg",
      "size": 53848,
      "modified": "2026-08-09T11:21:58.478Z"
    },
    {
      "name": "img_1786274542604-486245928.png",
      "url": "/uploads/img_1786274542604-486245928.png",
      "size": 80968,
      "modified": "2026-08-09T11:22:22.607Z"
    },
    {
      "name": "img_1786274603247-534870236.jpg",
      "url": "/uploads/img_1786274603247-534870236.jpg",
      "size": 9315,
      "modified": "2026-08-09T11:23:23.249Z"
    },
    {
      "name": "img_1786274625299-761688024.png",
      "url": "/uploads/img_1786274625299-761688024.png",
      "size": 61924,
      "modified": "2026-08-09T11:23:45.302Z"
    },
    {
      "name": "img_1786274644069-268105265.jpg",
      "url": "/uploads/img_1786274644069-268105265.jpg",
      "size": 61664,
      "modified": "2026-08-09T11:24:04.073Z"
    },
    {
      "name": "img_1786274727481-963741227.jpeg",
      "url": "/uploads/img_1786274727481-963741227.jpeg",
      "size": 53848,
      "modified": "2026-08-09T11:25:27.483Z"
    },
    {
      "name": "img_1786274755925-787582710.jpeg",
      "url": "/uploads/img_1786274755925-787582710.jpeg",
      "size": 53848,
      "modified": "2026-08-09T11:25:55.928Z"
    },
    {
      "name": "img_1786275718209-426473034.jpg",
      "url": "/uploads/img_1786275718209-426473034.jpg",
      "size": 11653,
      "modified": "2026-08-09T11:41:58.212Z"
    },
    {
      "name": "img_1786275925285-870846440.jpg",
      "url": "/uploads/img_1786275925285-870846440.jpg",
      "size": 11653,
      "modified": "2026-08-09T11:45:25.288Z"
    }
  ]
}
```

8)



9)

```
db.images.find()
{
  _id: ObjectId('6a784730e7f3616a55d42026'),
  processingId: 'test_001',
  originalFilename: 'test.jpg',
  storedFilename: 'test.jpg',
  filePath: './uploads/test.jpg',
  mimeType: 'image/jpeg',
  fileSize: 1024,
  width: 800,
  height: 600,
  format: 'jpeg',
  status: 'pending',
  createdAt: 2026-08-09T09:24:00.518Z,
  updatedAt: 2026-08-09T09:24:00.518Z
}
```